

Subject Description Form

Subject Code	COMP3011				
Subject Title	Design and Analysis of Algorithms				
Credit Value	3				
Level	3				
Pre-requisite / Co-requisite / Exclusion	Pre-requisite: COMP2011/COMP2013/DSAI2201/EIE3320 or equivalent				
Objectives	The objectives of this subject are to: 1. provide students with in-depth knowledge on algorithm design techniques; and 2. introduce and practice advanced algorithms for various data types.				
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) understand common techniques for designing algorithms; (b) acquire the skills to design efficient algorithms for solving computational problems; (c) analyse and compare the efficiency of algorithms; (d) design and implement efficient algorithms for solving computing problems in a high-level programming language (e.g., C++ or Java); <u>Attributes for all-roundedness</u> (e) solve problems independently; and (f) think critically for improvement in solutions.				
Subject Synopsis/ Indicative Syllabus	<table><tr><td>Topic</td></tr><tr><td>1. Analysis of algorithms Mathematical techniques; big-O notation; efficiency analysis; recurring relations.</td></tr><tr><td>2. Advanced Algorithmic Design Techniques Dynamic programming, divide-and-conquer, branch-and-bound, greedy algorithm.</td></tr><tr><td>3. Advanced Analysis Techniques Introduction to randomised algorithms, probabilistic analysis, amortised analysis.</td></tr></table>	Topic	1. Analysis of algorithms Mathematical techniques; big-O notation; efficiency analysis; recurring relations.	2. Advanced Algorithmic Design Techniques Dynamic programming, divide-and-conquer, branch-and-bound, greedy algorithm.	3. Advanced Analysis Techniques Introduction to randomised algorithms, probabilistic analysis, amortised analysis.
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	4. Computational Geometry Algorithms Spatial range searching, indexing of spatial objects, convex hull, closest pairs																																																										
	5. NP-Complete Problems Complexity classes, NP-completeness, reduction, approximation algorithms.																																																										
Teaching/ Learning Methodology	Lectures provide students the main concepts of the topic, together with comprehensive examples for easy understanding. Tutorials and lab sessions offer an opportunity to students for practicing their algorithmic analysis, design, and implementation techniques. Both written and programming assignments will be utilised in the course. Written assignments help students develop analysis and design skills, whereas programming assignments emphasise on implementation skills.																																																										
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighti ng</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Continuous Assessment</td><td rowspan="3">30%</td><td colspan="6"></td></tr><tr><td>1. Assignments</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Mid-Term / Tests</td><td>✓</td><td>✓</td><td>✓</td><td></td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>70%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table>	Specific assessment methods/tasks	% weighti ng	Intended subject learning outcomes to be assessed						a	b	c	d	e	f	Continuous Assessment	30%							1. Assignments	✓	✓	✓	✓	✓		2. Mid-Term / Tests	✓	✓	✓		✓	✓	Examination	70%	✓	✓	✓		✓	✓	Total	100%							Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: All three items are relevant to the assessment of the use of algorithms for problem solving, as well as their efficiency analysis (for items a, b, c). In addition, programming exercises in assignments are used to assess implementation skills (for item d); whereas the mid-term / tests and the examination are used to assess independent problem solving and critical thinking skills (for items e, f).					
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Reading List and References	Textbook: 1. Cormen, Thomas H., Leiserson, Charles E., Rivest, Ronald L. and Stein, Clifford, <i>Introduction to Algorithms</i>, 3rd Edition, MIT Press, 2009. Reference Books: 1. Goodrich, M.T., and Tamassia, R., <i>Data Structures and Algorithms in Java</i>, 3rd Edition, John Wiley, 2005. 2. Carrano, Frank M., <i>Data Abstraction & Problem Solving with C++: Walls & Mirrors</i>, Addison Wesley, 2007. 3. Jon M. Kleinberg, Éva Tardos: <i>Algorithm design</i>. Addison-Wesley 2006, ISBN 978-0-321-37291-8.
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